CM1

Set systems and graphs, the experts problem, and an introduction to discrepancy

What we will do today

1. Basic combinatorics: set systems and graphs.
2. The experts problem: how to predict well when one expert is always right.
3. The discrepancy problem: coloring points so that every set is balanced.

Set systems

Definition 1 (Set system)

Let X be a finite set, and let $\mathscr{R}$ be a set of subsets of X. We call $(X, \mathscr{R})$ a *set system* with *ground set* X.

Examples of set systems

- Intervals. $X = \{1, \ldots, n\}$, points on a line;

$$\mathscr{R} \;=\; \big\{\, \{a, a+1, \ldots, b\} \;:\; 1 \le a \le b \le n \,\big\},$$

  the family of all intervals.
- Graphs. $X = V$, the vertex set of a graph, and $\mathscr{R} = E$: each edge is a set of size 2.
- Geometry. $X =$ a set of points in the plane, and $\mathscr{R} =$ the subsets of the form $X \cap H$, where H is a halfplane (the points on one side of a line).

Graphs as set systems

Definition 2 (Graph)

A *graph* is a pair $G = (V, E)$ where V is a finite set of *vertices* and $E \subseteq \binom{V}{2}$ is a set of *edges* (2-element subsets of V).

- So a graph *is* a set system: vertex set V, and every edge is a set of size 2.
- A graph on n vertices has at most $\binom{n}{2}$ edges.
- Two-colorings of graphs will reappear below: coloring vertices ± 1 so that no edge is monochromatic.

Two-colorings

Definition 3 (Two-coloring)

A *two-coloring* of X is an assignment of a -1 or a $+1$ color to each element of X:

$$\chi \colon X \to \{-1, +1\}.$$

- There are exactly $2^{|X|}$ two-colorings of X: each element gets one of two colors, with no restriction between different elements.
- We think of $\chi(v) = +1$ as "red" and $\chi(v) = -1$ as "blue".

Discrepancy: measuring balance

Definition 4 (Discrepancy)

Given a set system $(X, \mathscr{R})$ and a two-coloring χ, the *discrepancy* of a set $S \subseteq X$ is

$$\text{disc}_\chi(S) = \left| \sum_{v \in S} \chi(v) \right|,$$

and the *discrepancy* of $\mathscr{R}$ with respect to χ is

$$\text{disc}_\chi(\mathscr{R}) = \max_{S \in \mathscr{R}} \text{disc}_\chi(S).$$

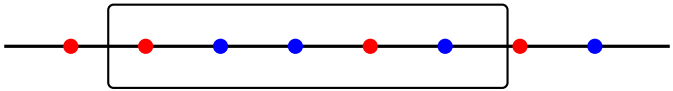
The discrepancy of S measures the "balance" of the colors of the elements of S.

Our goal: find a two-coloring χ that minimizes $\text{disc}_\chi(\mathscr{R})$.

A set with small discrepancy is balanced

$\sum_{v \in S} \chi(v) = 2 - 3 = -1$ (nearly balanced)

S: two red, three blue



If $\mathrm{disc}_\chi(S)$ is small, then S has roughly the same number of red and blue elements.

The easy case: a single set

Remark 5 (Trivial)

It is easy to find a coloring that minimizes the discrepancy of a single set S: color the elements of S as evenly as possible between -1 and $+1$.

Then

$$\operatorname{disc}_\chi(S) = \begin{cases} 0 & \text{if } |S| \text{ is even,} \\ 1 & \text{if } |S| \text{ is odd.} \end{cases}$$

The challenge lies in finding a *single* coloring such that *all* sets of $\mathscr{R}$ have low discrepancy. This is much harder, because the different constraints may conflict with each other.

Example: discrepancy of graphs

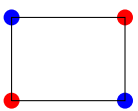
Let $G = (V, E)$ be a graph, seen as a set system. For an edge $\{u, v\} \in E$:

$$\mathrm{disc}_\chi(\{u, v\}) \;=\; |\chi(u) + \chi(v)| \;=\; \begin{cases} 0 & \text{if } u, v \text{ have different colors,} \\ 2 & \text{if } u, v \text{ have the same color.} \end{cases}$$

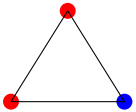
Hence

$$\mathrm{disc}_\chi(G) = \begin{cases} 0 & \text{if no edge is monochromatic,} \\ 2 & \text{otherwise.} \end{cases}$$

Graphs with discrepancy 0: bipartite graphs



C_4: discrepancy 0



C_3: discrepancy 2

Proposition 6

A graph has a coloring χ with $\mathrm{disc}_\chi(G) = 0$ iff it is bipartite.

Proof.

If $\mathrm{disc}_\chi(G) = 0$, every edge joins two different colors, so the two color classes form a bipartition. Conversely, if G is bipartite, color the two parts differently; then every edge has discrepancy 0. In particular, no odd cycle admits such a coloring: some edge is monochromatic, giving discrepancy 2. □

The experts problem

Definition 7 (Experts problem)

There are n experts, indexed by $[n]$, where $[n] = \{1, \ldots, n\}$, and m events, indexed by $t \in [m]$. Each event has a binary outcome, $+1$ or -1.

At iteration t:

1. each expert $i \in [n]$ makes a prediction for event t;
2. after seeing all predictions, *we* make a prediction;
3. the true outcome of event t is revealed.

- $\mathbf{1}_i^t = 1$ iff expert i predicted event t *incorrectly*; otherwise $\mathbf{1}_i^t = 0$.
- $\mathbf{1}_{\text{us}}^t = 1$ iff *we* predicted event t incorrectly; otherwise $\mathbf{1}_{\text{us}}^t = 0$.

Our goal

We would like to upper bound our total number of mistakes

$$\sum_{t=1}^{m} \mathbf{1}_{\text{us}}^t \qquad \text{in terms of} \qquad \sum_{t=1}^{m} \mathbf{1}_{i^*}^t, \qquad \text{where} \qquad i^* = \arg\min_{i \in [n]} \sum_{t=1}^{m} \mathbf{1}_i^t.$$

In words: we want to make almost as few mistakes as the *best* expert — who we only know *in hindsight*!

No assumptions: the experts may conspire against us; the only requirement is that we see all their predictions *before* we decide.

A warm-up: is following the majority good enough?

- Suppose one expert is *always* correct, and we simply follow the majority prediction (by count).
- If the side we follow is wrong, then at least half of the experts were wrong — strictly more than half, unless there is a tie — and we may discard all the wrong ones.

Theorem 8

If some expert is always correct, then there is a deterministic algorithm making at most $\lfloor \log_2 n \rfloor$ *mistakes in total.*

$\lfloor \log_2 n \rfloor$ errors — regardless of the number m of events!

Algorithm: majority and discard

Algorithm (majority and discard)

Keep a set of *alive* experts; initially all n experts are alive. For $t = 1, \ldots, m$:

1. each alive expert makes a prediction for event t;
2. we predict the outcome of the majority (by count) of the alive experts (ties broken arbitrarily);
3. the true outcome is revealed;
4. we discard every alive expert who was wrong.

- The always-correct expert is *never* discarded.
- Whenever we err, the side we followed was wrong $\Rightarrow$ at least half of the alive experts are discarded.

Analysis: why at most $\lfloor \log_2 n \rfloor$ mistakes?

Proof.

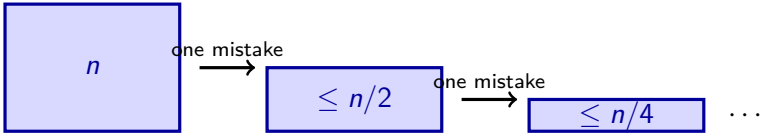
- Let k_t = number of alive experts at the start of iteration t.
- If we err at iteration t, the side we followed was wrong and it had $\geq k_t/2$ experts; all of them are discarded, so
$$k_{t+1} \leq \frac{k_t}{2}.$$
- If we are correct, $k_{t+1} \leq k_t$ (we never gain experts).
- The always-correct expert survives to the end, so $k_t \geq 1$ always.

Starting from $k_1 = n$: each mistake at least halves the count, and the count never reaches 0. Hence the number of mistakes is at most

$$\lfloor \log_2 n \rfloor.$$



Picture: each mistake halves the alive experts



$$\frac{n}{2^k} \geq 1 \qquad \Longleftrightarrow \qquad k \leq \log_2 n,$$

so for integer k,

$$k \leq \lfloor \log_2 n \rfloor.$$

Example 9 ($n = 8$)

At most $\lfloor \log_2 8 \rfloor = 3$ mistakes: $8 \to 4 \to 2 \to 1$, and the one surviving expert is the always-correct one.

Is the $\log_2 n$ bound optimal? An adversary says yes

Proposition 10 (Lower bound)

For any deterministic algorithm, an adversary can choose the experts' predictions and the outcomes so that the algorithm makes at least $\lfloor \log_2 n \rfloor$ mistakes, while some expert makes none.

Proof.

- Split the *clean* experts (those with no mistake so far) evenly into two groups, one predicting $+1$, one predicting -1 — possible while at least two clean experts remain.
- After our prediction, reveal the outcome that makes *us* wrong; the group that predicted it stays clean.

After k rounds at least $\lfloor n/2^k \rfloor$ clean experts remain; while two remain (all $k \leq \lfloor \log_2 n \rfloor - 1$) we force another mistake: at least $\lfloor \log_2 n \rfloor$ mistakes, with one expert still clean. □

Check yourself

Quick question: Suppose there is *no* expert who is always correct. Is the majority-and-discard algorithm still guaranteed to work?

No! It can discard *all* experts and run out of advice.

(If no expert is always correct, one can restart the algorithm every time the alive list becomes empty. If M is the best expert's number of mistakes, this gives $O((M+1)\log n)$ mistakes. We will see this and better algorithms next time.)

The discrepancy problem

- Given $(X, \mathscr{R})$, find a two-coloring $\chi : X \to \{-1, +1\}$ that minimizes

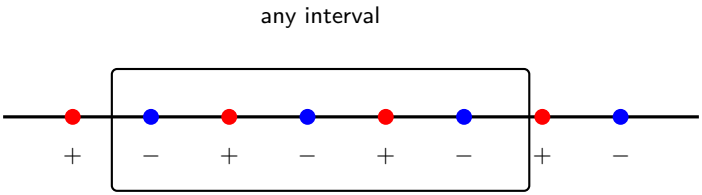
$$\mathrm{disc}_\chi(\mathscr{R}) \;=\; \max_{S \in \mathscr{R}} \Big| \sum_{v \in S} \chi(v) \Big|.$$

- One set is easy; *all* sets simultaneously is hard: the constraints may conflict.

Let us start with the most natural geometric example:

$X =$ points on a line, $\quad \mathscr{R} =$ all intervals.

Geometric example: intervals on a line



Idea: color the points alternately $+1, -1, +1, -1, \ldots$ from left to right.

Theorem 11 (Intervals are easy)

The alternating coloring has discrepancy at most 1 *on every interval.*

Why every interval is balanced

Proof.

Any interval contains *consecutive* points of the line. In the alternating coloring these are

$$+1, -1, +1, -1, \ldots$$

so their sum is:

- 0 if the interval has an even number of points,
- $+1$ or -1 if it has an odd number of points.

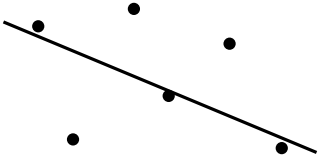
Therefore every interval S has

$$\left|\sum_{v \in S} \chi(v)\right| \in \{0, 1\},$$

so $\mathrm{disc}_\chi(\mathscr{R}) \leq 1$ — the smallest possible value, since a one-point interval always contributes 1. □

A slightly harder geometric example

each line splits the points into two sides



- Now the sets are all subsets of the form $X \cap H$, where H is a halfplane (one side of a line).
- The optimal discrepancy is no longer $O(1)$: the optimal worst-case halfplane discrepancy for n points in the plane is

$$\Theta\left(n^{1/4}\right).$$

The structure of the set system matters enormously: intervals give $O(1)$, halfplanes give $\Theta(n^{1/4})$.

Summary of today

Definitions

- Set system $(X, \mathscr{R})$; graphs are set systems with edges of size 2.
- Two-coloring $\chi : X \to \{-1, +1\}$; discrepancy $\mathrm{disc}_\chi(S) = \left|\sum_{v \in S} \chi(v)\right|$, $\mathrm{disc}_\chi(\mathscr{R}) = \max_{S \in \mathscr{R}} \mathrm{disc}_\chi(S)$.

Experts

If one expert is always correct: majority-and-discard $\Rightarrow$ at most $\lfloor \log_2 n \rfloor$ mistakes, and this bound is optimal for deterministic algorithms.

Discrepancy

- Intervals on a line: alternating coloring $\Rightarrow \leq 1$; halfplanes in the plane: $\Theta(n^{1/4})$.
- Any system with $m = |\mathscr{R}|$ sets: a random coloring gives $O(\sqrt{n \ln m})$ with probability $\geq 1 - 1/m$.

Next class

1. A deterministic algorithm for discrepancy $O(\sqrt{n \ln m})$ — the multiplicative weights technique, already glimpsed in the experts problem.
2. The general experts problem without a perfect expert.